UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. Which data type stores a whole number?
   A. string
   B. float
   C. char
   D. int

2. How many times does a while loop execute when its condition is false initially?
   A. Two
   B. Until the user stops it
   C. Zero
   D. One

3. What is pseudocode?
   A. A C++ header file
   B. Machine code produced by an assembler
   C. A structured English-like description of an algorithm
   D. A type of compiler error

4. Which part of a for loop is executed after each iteration?
   A. The condition only
   B. The update expression
   C. The initialisation
   D. The loop body twice

5. Which statement immediately exits a loop?
   A. return 0
   B. cin
   C. continue
   D. break

6. Which declaration creates an integer array of five elements?
   A. int a(5);
   B. array a[5];
   C. int a[5];
   D. int a;

7. Which input pattern stops when the user enters -1?
   A. A fixed for loop
   B. A sentinel-controlled loop
   C. A function prototype
   D. A switch statement

8. What does a compiler do?
   A. Translates the whole source program before execution
   B. Draws flowcharts automatically
   C. Stores only integer values
   D. Executes one source statement at a time

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9. What does the % operator return?
   A. The square root
   B. The absolute value
   C. The quotient
   D. The remainder of integer division

10. Which keyword prevents a variable's value from being changed after initialisation?
   A. const
   B. static
   C. void
   D. break

PART B (60 MARKS)

QUESTION 1

This question tests the program development life cycle.

a) Define source code and object code.

(3 marks)

b) Describe THREE stages of the program development life cycle.

(2 marks)

QUESTION 2

A program reads three integer values and displays the largest value.

a) Draw a flowchart to find the largest of three values.

(6 marks)

b) Write pseudocode and a complete C++ program for the flowchart.

(7 marks)

QUESTION 3

This question tests type conversion and constants.

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

(5 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

(6 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
   s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(10 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(5 marks)

QUESTION 5

A class records ten student marks in an array.

a) Declare an integer array of ten marks and read the marks from the user.

(8 marks)

b) Write a program segment to calculate and display the average, highest mark and number of marks above the average.

(8 marks)

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PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

END OF QUESTION PAPER